

# Status Report Number 1

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## 1 Architecture and intervention ladder

Figure 1 locates the activation sites and organizes the progressive experimental program.

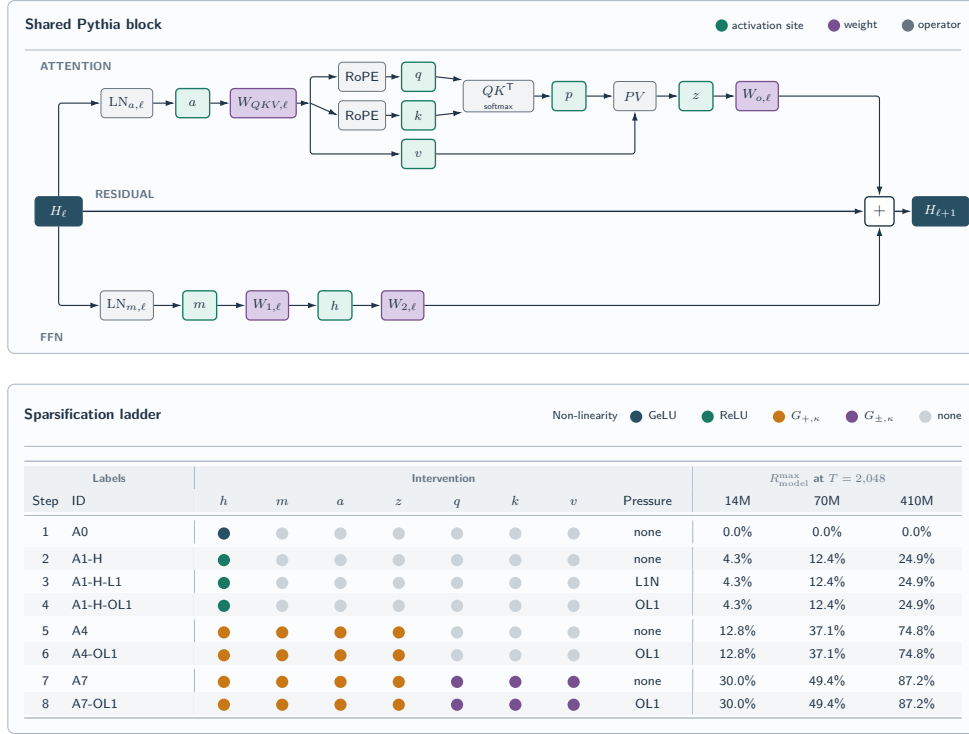


Figure 1: Shared Pythia block and sparsification intervention ladder. A4 and A7 denote operational topologies A4-Z and A7-Z-POST.

## 2 Experiment control

### 2.1 Scope dashboard

**Pilot.** Local execution with 581 optimizer boundaries, global batch 64, and 76,152,832 training input tokens per condition.

**Paper-scale full pass.** RunPod A100 execution with 712 optimizer boundaries, global batch 1,024, and 1,493,172,224 training input tokens per condition. Conditions run concurrently across independent GPUs.

Table 1: Experiment scope at the report cutoff.

| Step     | Conditions/model | Pythia-14M                  | 70M | 410M |
|----------|------------------|-----------------------------|-----|------|
| A0       | 1                | Full pass complete: Run 004 | 0/1 | 0/1  |
| A1-H     | 1                | Full pass complete: Run 004 | 0/1 | 0/1  |
| A1-H-L1  | 4                | 4/4 full pass: Run 004      | 0/4 | 0/4  |
| A1-H-OL1 | 4                | 4/4 full pass: Run 009      | 0/4 | 0/4  |
| A4       | 5                | 5/5 full pass: Run 011      | 0/5 | 0/5  |
| A4-OL1   | 5                | 5/5 full pass: Run 012      | 0/5 | 0/5  |
| A7       | 5                | 5/5 full pass: Run 013      | 0/5 | 0/5  |
| A7-OL1   | 5                | 5/5 full pass: Run 014      | 0/5 | 0/5  |

## 2.2 Paper-scale contract and parameter set

The scale plan now adopts the size-specific learning rates in the standard Pythia recipe: peak/minimum  $10^{-3}/10^{-4}$  for 14M and 70M, and  $3 \times 10^{-4}/3 \times 10^{-5}$  for 410M. The study applies each activation intervention within this established recipe.

Table 2: Shared paper-scale execution contract.

| Field              | Specification                                                                                            |
|--------------------|----------------------------------------------------------------------------------------------------------|
| Model construction | Random initialization from pinned Pythia architecture; model seed 1234                                   |
| Data               | Pinned MiniPile cache; data-order seed 1234; sequence length 2,048                                       |
| Training horizon   | 712 optimizer boundaries; 1,493,172,224 input tokens per condition                                       |
| Global batch       | 1,024 sequences; hardware-specific microbatch and accumulation preserve the product                      |
| Optimizer          | AdamW, betas (0.9, 0.95), epsilon $10^{-8}$ , weight decay 0.1, global clipping 1.0                      |
| Schedule           | 1% linear warmup; cosine decay to 10% of the size-specific peak learning rate                            |
| Precision          | Dynamic FP16 autocast; FP32 parameters and optimizer state                                               |
| Validation         | 500 documents; 338 complete blocks; 692,224 input tokens; 1,444-token tail recorded                      |
| Retention          | Activation counts and moments, weight norms, logical counts, pressure-gradient metrics, final checkpoint |

Table 3: Aggregated intervention parameter set. Threshold equality survives.

| Family   | Gate and pressure                                | Parameter grid                                 |
|----------|--------------------------------------------------|------------------------------------------------|
| A0       | Stock GeLU at <b>h</b>                           | Singleton                                      |
| A1-H     | ReLU at <b>h</b>                                 | Singleton                                      |
| A1-H-L1  | ReLU; naive L1 at <b>h</b>                       | $\lambda \in \{0.05, 0.1, 0.5, 1.0\}$          |
| A1-H-OL1 | ReLU; OL1 at <b>h</b>                            | Same $\lambda$ grid; budget 1.0                |
| A4       | One-sided threshold at <b>a, m, h, z</b>         | $\kappa \in \{0, 0.01, 0.05, 0.1, 0.5\}$       |
| A4-OL1   | A4 gates; OL1 at four sites                      | Same $\kappa$ grid; $\lambda = 1$ ; budget 1.0 |
| A7       | A4 plus symmetric <b>q_post, k_post, v</b> gates | Same $\kappa$ grid                             |
| A7-OL1   | A7 gates; OL1 at seven sites                     | Same $\kappa$ grid; $\lambda = 1$ ; budget 1.0 |

## 3 RunPod compute

Table 4 reports elapsed wall-clock time for each condition’s scientific process. Effective recorded RunPod cost across the six runs is \$89.99; the retained 100 GB network volume remains separate at \$7/month.

Table 4: RunPod execution record. Wall-clock is elapsed scientific-process time per completed condition.

| Run          | Compute                         | Wall-clock | Cost           |
|--------------|---------------------------------|------------|----------------|
| 004          | 5× A100 80 GB; six conditions   | 63–108 min | \$21.52        |
| 009          | 4× A100 80 GB; four conditions  | 86–105 min | \$12.27        |
| 011          | 5× A100 80 GB; five conditions  | 59–104 min | \$11.47        |
| 012          | 5× A100 80 GB; five conditions  | 70–100 min | \$14.77        |
| 013          | 5× A100 80 GB; five conditions  | 82–89 min  | \$15.83        |
| 014          | 5× A100 80 GB; five conditions  | 74–115 min | \$14.13        |
| <b>Total</b> | <b>Effective recorded spend</b> | —          | <b>\$89.99</b> |

Deployment required an 80 GB memory class, cache staging, detached execution, resumable artifact transfer, hash verification, and explicit teardown. The final Run 014 audit found zero Pods; the pre-existing retained volume was unchanged.

## 4 Results

The analysis adds one intervention at each stage and maps the resulting validation-loss–logical-compute frontier. Only completed comparisons appear as result subsections.

All site columns report exact-zero percentages; **q** and **k** denote the post-RoPE tensors. “n.m.” denotes a site that was not measured in the source run.

Table 5: Progressive analysis status.

| Stage   | Comparison                          | Evidence status                                           |
|---------|-------------------------------------|-----------------------------------------------------------|
| 1       | A0 → A1-H                           | Full-pass endpoints complete                              |
| Control | A0/A1-H → uniform post-hoc clipping | Full-pass Analysis 005 complete                           |
| 2       | A1-H → A1-H-L1                      | Full-pass dose response complete                          |
| 3       | A1-H-L1 → A1-H-OL1                  | Matched full-pass Analysis 003 complete                   |
| 4       | A1-H → A4                           | Matched full-pass Analysis 004 complete                   |
| 5       | A4 → A4-OL1                         | Matched full-pass Analysis 007; tentative F001            |
| 6       | A4 → A7                             | Matched full-pass Analysis 008; tentative F002            |
| 7       | A7 → A7-OL1                         | Five matched full-pass pairs in Analysis 008; descriptive |

### 4.1 Type of non-linearity: A0 to A1-H

ReLU creates a 63.448% exact-zero rate at **h** and moves  $R_{\text{model}}$  to 2.714%.

Table 6: Pythia-14M full-pass controls from Run 004.

| Condition  | Loss  | $R_{\text{model}}$ (%) | Exact-zero (%) |          |          |          |          |          |          |
|------------|-------|------------------------|----------------|----------|----------|----------|----------|----------|----------|
|            |       |                        | <b>h</b>       | <b>m</b> | <b>a</b> | <b>z</b> | <b>q</b> | <b>k</b> | <b>v</b> |
| A0: GeLU   | 5.209 | < 0.001                | < 0.001        | 0.000    | n.m.     | n.m.     | < 0.001  | < 0.001  | < 0.001  |
| A1-H: ReLU | 5.270 | 2.714                  | 63.448         | 0.000    | n.m.     | n.m.     | < 0.001  | < 0.001  | < 0.001  |

The evaluation-only uniform TEAL-style control thresholds **a**, **m**, **h**, and **z** separately in every block; it does not threshold **q**, **k**, or **v**. For one common target  $p$ , values satisfying  $|x| \leq t$  are zeroed without retraining. Table 7 is the only result table that reports paired loss deltas.

Table 7: Selected uniform TEAL-style post-hoc endpoints from Analysis 005, Observation O001. Deltas are paired to the independently evaluated  $p = 0$  row for the same checkpoint.

| $p$ | GeLU control |                     |                        | ReLU control |                     |                        |
|-----|--------------|---------------------|------------------------|--------------|---------------------|------------------------|
|     | Loss         | $\Delta\text{loss}$ | $R_{\text{model}}$ (%) | Loss         | $\Delta\text{loss}$ | $R_{\text{model}}$ (%) |
| 0.0 | 5.208594     | 0.000000            | < 0.001                | 5.269633     | 0.000000            | 2.714133               |
| 0.1 | 5.214453     | 0.005859            | 1.278706               | 5.274073     | 0.004439            | 3.568403               |
| 0.2 | 5.250270     | 0.041676            | 2.554786               | 5.307940     | 0.038307            | 4.418365               |
| 0.3 | 5.361901     | 0.153307            | 3.831641               | 5.405056     | 0.135422            | 5.271871               |
| 0.4 | 5.605384     | 0.396790            | 5.125544               | 5.602965     | 0.333331            | 6.129325               |
| 0.5 | 6.077731     | 0.869137            | 6.441923               | 5.976274     | 0.706641            | 6.994451               |

## 4.2 Naive L1 pressure: A1-H to A1-H-L1

Increasing  $\lambda$  raises exact-zero mass and  $R_{\text{model}}$  across the four full-pass endpoints.

Table 8: Run 004 full-pass naive-L1 dose response.

| $\lambda$ | Loss  | $R_{\text{model}}$ (%) | Exact-zero (%) |          |          |          |          |          |          |
|-----------|-------|------------------------|----------------|----------|----------|----------|----------|----------|----------|
|           |       |                        | <b>h</b>       | <b>m</b> | <b>a</b> | <b>z</b> | <b>q</b> | <b>k</b> | <b>v</b> |
| 0.05      | 5.206 | 3.142                  | 73.441         | 0.000    | n.m.     | n.m.     | < 0.001  | < 0.001  | < 0.001  |
| 0.10      | 5.165 | 3.336                  | 77.993         | 0.000    | n.m.     | n.m.     | < 0.001  | < 0.001  | < 0.001  |
| 0.50      | 5.113 | 3.788                  | 88.542         | 0.000    | n.m.     | n.m.     | < 0.001  | < 0.001  | < 0.001  |
| 1.00      | 5.102 | 3.949                  | 92.323         | 0.000    | n.m.     | n.m.     | < 0.001  | < 0.001  | < 0.001  |

## 4.3 Orthogonalization: A1-H-L1 to A1-H-OL1

The matched full-pass endpoints occupy a narrow common region in validation loss,  $R_{\text{model}}$ , and **h** sparsity.

Table 9: Absolute naive-L1 and OL1 endpoints from Analysis 003.

| $\lambda$ | Method | Loss  | $R_{\text{model}}$ (%) | Exact-zero (%) |          |          |          |          |          |          |
|-----------|--------|-------|------------------------|----------------|----------|----------|----------|----------|----------|----------|
|           |        |       |                        | <b>h</b>       | <b>m</b> | <b>a</b> | <b>z</b> | <b>q</b> | <b>k</b> | <b>v</b> |
| 0.05      | L1     | 5.206 | 3.142                  | 73.441         | 0.000    | n.m.     | n.m.     | < 0.001  | < 0.001  | < 0.001  |
|           | OL1    | 5.198 | 3.145                  | 73.526         | 0.000    | n.m.     | n.m.     | < 0.001  | < 0.001  | < 0.001  |
| 0.10      | L1     | 5.165 | 3.336                  | 77.993         | 0.000    | n.m.     | n.m.     | < 0.001  | < 0.001  | < 0.001  |
|           | OL1    | 5.159 | 3.339                  | 78.046         | 0.000    | n.m.     | n.m.     | < 0.001  | < 0.001  | < 0.001  |
| 0.50      | L1     | 5.113 | 3.788                  | 88.542         | 0.000    | n.m.     | n.m.     | < 0.001  | < 0.001  | < 0.001  |
|           | OL1    | 5.110 | 3.768                  | 88.084         | 0.000    | n.m.     | n.m.     | < 0.001  | < 0.001  | < 0.001  |
| 1.00      | L1     | 5.102 | 3.949                  | 92.323         | 0.000    | n.m.     | n.m.     | < 0.001  | < 0.001  | < 0.001  |
|           | OL1    | 5.121 | 3.938                  | 92.067         | 0.000    | n.m.     | n.m.     | < 0.001  | < 0.001  | < 0.001  |

Figure 2 isolates the OL1 Adam-relative geometry before and after conflict projection.

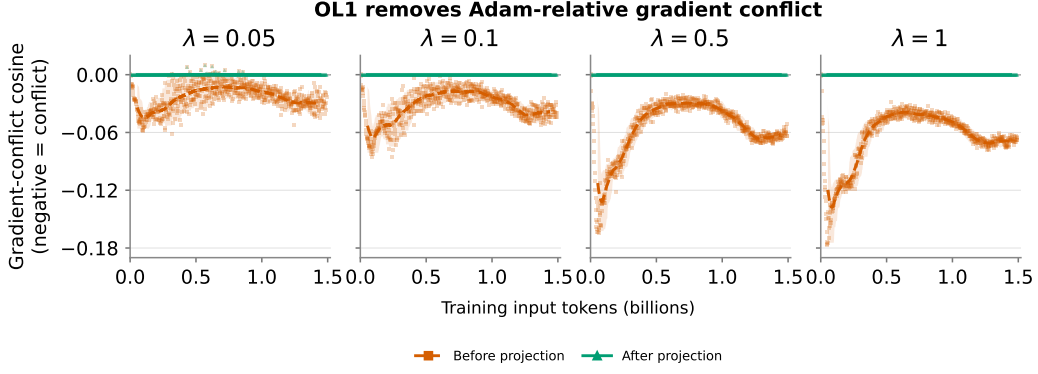


Figure 2: OL1 Adam-relative interaction before and after conflict projection. Each panel contains 712 optimizer boundaries for one  $\lambda$ ; lines are centered 51-boundary means and bands span p05–p95. Source: Analysis 003, Observation O004.

#### 4.4 Architecture-wide thresholding and OL1: A4 to A4-OL1

A4 applies one-sided thresholds at **a**, **m**, **h**, and **z**; A4-OL1 adds four-site OL1 pressure with  $\lambda = 1$  and trust budget 1.0. A4-OL1 improves loss and  $R_{\text{model}}$  at every matched  $\kappa \leq 0.1$ , supporting tentative Finding F001; at  $\kappa = 0.5$ , the loss advantage reverses.

Table 10: Absolute A4 and A4-OL1 full-pass endpoints from Analysis 007, Observation O001.

| $\kappa$ | Method | Loss  | $R_{\text{model}}$ (%) | Exact-zero (%) |          |          |          |          |          |          |
|----------|--------|-------|------------------------|----------------|----------|----------|----------|----------|----------|----------|
|          |        |       |                        | <b>h</b>       | <b>m</b> | <b>a</b> | <b>z</b> | <b>q</b> | <b>k</b> | <b>v</b> |
| 0.00     | A4     | 5.470 | 7.212                  | 68.765         | 49.779   | 48.560   | 54.525   | < 0.001  | < 0.001  | < 0.001  |
|          | A4-OL1 | 5.216 | 8.409                  | 93.543         | 49.816   | 49.413   | 64.671   | < 0.001  | < 0.001  | < 0.001  |
| 0.01     | A4     | 5.466 | 7.414                  | 71.181         | 50.397   | 49.095   | 59.643   | < 0.001  | < 0.001  | < 0.001  |
|          | A4-OL1 | 5.205 | 8.586                  | 94.871         | 50.390   | 49.942   | 71.971   | < 0.001  | < 0.001  | < 0.001  |
| 0.05     | A4     | 5.434 | 8.206                  | 81.340         | 52.712   | 51.155   | 77.641   | < 0.001  | < 0.001  | < 0.001  |
|          | A4-OL1 | 5.196 | 9.036                  | 97.915         | 52.391   | 51.791   | 88.307   | < 0.001  | < 0.001  | < 0.001  |
| 0.10     | A4     | 5.420 | 8.953                  | 91.853         | 55.200   | 53.244   | 89.234   | < 0.001  | < 0.001  | < 0.001  |
|          | A4-OL1 | 5.229 | 9.343                  | 99.125         | 54.747   | 53.809   | 96.774   | < 0.001  | < 0.001  | < 0.001  |
| 0.50     | A4     | 5.660 | 10.216                 | 99.876         | 66.386   | 63.434   | 99.881   | < 0.001  | < 0.001  | < 0.001  |
|          | A4-OL1 | 5.723 | 10.227                 | 99.944         | 66.441   | 63.622   | 99.934   | < 0.001  | < 0.001  | < 0.001  |

## 4.5 Attention-operand thresholding and OL1: A4 to A7 to A7-OL1

A7 adds symmetric thresholds at post-RoPE  $q$ ,  $k$ , and  $v$  to A4’s one-sided  $a$ ,  $m$ ,  $h$ , and  $z$  gates; the matched A4/A7 result supports tentative Finding F002. A7-OL1 adds post-gate orthogonal L1 pressure at all seven active sites with  $\lambda = 1$  and trust budget 1. At matched  $\kappa = 0.1$ , A7-OL1 adds 1.3717 percentage points of  $R_{\text{model}}$  for +0.000816 validation loss; at  $\kappa = 0.5$ , it adds 12.0959 points for +0.126466 loss. The zero-threshold row regresses, so the A7/A7-OL1 effect is threshold-dependent and remains descriptive.

Table 11: Matched A7 and A7-OL1 full-pass endpoints from Analysis 008, Observation O001. Every percentage is recomputed after pooling integer counts over all six layers and 338 complete validation blocks. `out` is the ungated post- $W_o$  `attention_output`.

| $\kappa$ | Variant | Loss     | $R_{\text{model}}$ (%) | Exact-zero mass (%) |        |        |        |         |         |         |                  |
|----------|---------|----------|------------------------|---------------------|--------|--------|--------|---------|---------|---------|------------------|
|          |         |          |                        | $h$                 | $m$    | $a$    | $z$    | $q$     | $k$     | $v$     | <code>out</code> |
| 0.00     | A7      | 5.468401 | 7.217651               | 68.689              | 49.790 | 48.590 | 55.221 | < 0.001 | < 0.001 | < 0.001 | < 0.001          |
|          | A7-OL1  | 5.480184 | 7.054229               | 68.277              | 49.503 | 48.583 | 42.761 | < 0.001 | < 0.001 | < 0.001 | < 0.001          |
| 0.01     | A7      | 5.458822 | 7.617649               | 71.275              | 50.330 | 49.064 | 59.041 | 0.399   | 0.407   | 1.654   | < 0.001          |
|          | A7-OL1  | 5.475797 | 7.723393               | 71.050              | 50.103 | 49.196 | 48.933 | 1.373   | 1.496   | 2.338   | < 0.001          |
| 0.05     | A7      | 5.437888 | 9.126924               | 81.298              | 52.681 | 51.036 | 78.018 | 1.840   | 1.716   | 7.312   | < 0.001          |
|          | A7-OL1  | 5.462811 | 9.863032               | 81.052              | 52.241 | 51.036 | 72.067 | 5.427   | 5.817   | 10.526  | < 0.001          |
| 0.10     | A7      | 5.428681 | 10.425018              | 91.353              | 55.075 | 53.052 | 90.052 | 3.150   | 3.227   | 11.355  | < 0.001          |
|          | A7-OL1  | 5.429497 | 11.796760              | 91.607              | 54.874 | 53.104 | 88.896 | 8.227   | 8.544   | 18.819  | < 0.001          |
| 0.50     | A7      | 5.702923 | 15.386813              | 99.859              | 65.721 | 62.306 | 99.865 | 17.435  | 18.134  | 31.255  | < 0.001          |
|          | A7-OL1  | 5.829390 | 27.482684              | 99.876              | 68.698 | 75.732 | 99.918 | 93.545  | 94.541  | 98.713  | < 0.001          |

## 4.6 Trained and post-hoc full-pass frontier

Figure 3 places the trained endpoints and uniform TEAL-style clipping of the Run 004 controls on one validation-loss- $R_{\text{model}}$  plane. It includes all five A7 and five A7-OL1 full-pass endpoints; Table 11 supplies their matched thresholds and count-pooled site values.

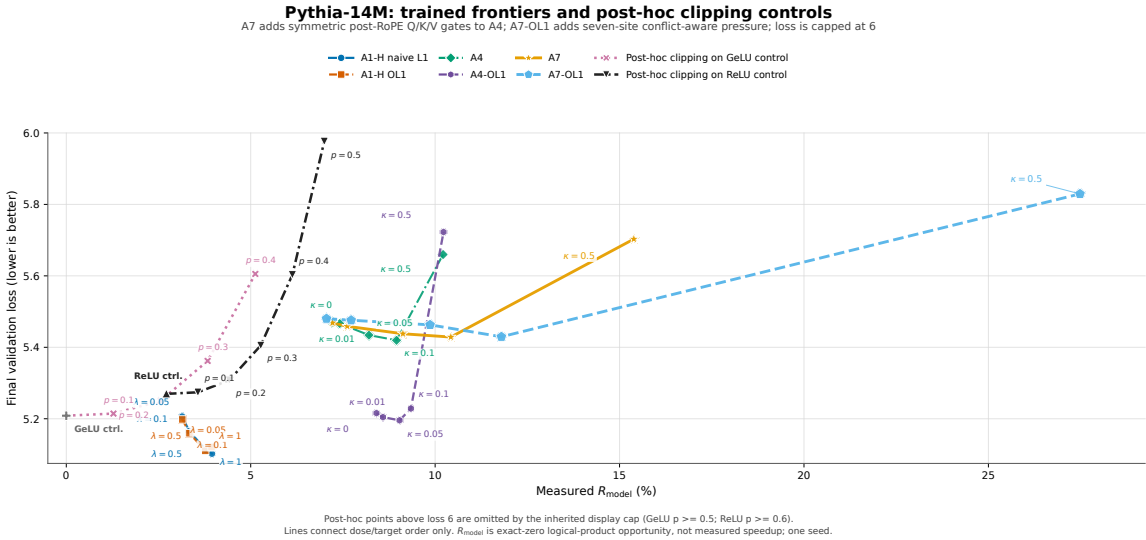


Figure 3: Trained full-pass endpoints, uniform TEAL-style post-hoc clipping of the Run 004 controls, and the matched Run 013 A7 and Run 014 A7-OL1 endpoints. A7 uses one-sided thresholds at  $a, m, h, z$  and symmetric thresholds at post-RoPE  $q, k, v$ ; A7-OL1 adds seven-site orthogonal L1 pressure. The vertical display is capped at loss 6; above-cap post-hoc points remain in the complete source sweep. Lines indicate dose or target order only. Source: Analysis 008, Observation O001.